Mark schemes

Q1.

(a) The boat is not affected by the movement of the water.

First assumption

B1

The resistance force will directly oppose the motion of the boat and be the only force that needs to be considered.

second assumption

B1

2

(b) The boat is a particle. / There is no wind. / No air resistance.

Appropriate assumption

B1

1

(c) (i) $80\frac{dv}{dt} = -20v$

Use of Newton's second law, $\frac{dv}{dt}$ *and 20v.*

M1

$\frac{dv}{dt} = -\frac{v}{4}$

Correct result

A1

2

(ii) $\int \frac{1}{v}\,dv = -\int \frac{1}{4}\,dt$

Sep. of variables and forming integrals

M1

$\ln v = -\frac{t}{4} + c$

Integrating to get an 1n v term

dM1

Correct integrals with or without c

A1

$v = Ae^{-\frac{t}{4}}$

$v = 12$, $t = 0 \Rightarrow A = 12$

Finding A or c

dM1

$v = Ae^{-\frac{t}{4}}$

Correct final result

A1

5

[10]

Q2.

(a) $20\frac{dv}{dt} = -10\sqrt{v}$

applying Newton's second law with $\frac{dv}{dt}$

M1

correct differential equation

A1

$\frac{dv}{dt} = -\frac{\sqrt{v}}{2}$

separating variables

dM1

$\int \frac{1}{\sqrt{v}}\,dv = \int -\frac{1}{2}\,dt$ AG

$2\sqrt{v} = \frac{t}{2} + c$

integrating

dM1

correct integrals

A1

$t = 0$, $v = 25 \Rightarrow c = 10$

finding the constant of integration

$v = \left(\frac{20-t}{4}\right)^2$

correct final result from correct working

A1

7

(b) $t = 20$

correct time

B1

1

[8]

Q3.

$\frac{dv}{dt} = \frac{k}{v}$

B1

$\int v\,dv = \int k\,dt$

Separation of variables involving t

M1

$\frac{v^2}{2} = kt + c$

m1

Integrate

A1

$t = 0,\ v = u,\ \therefore\ c = \frac{u^2}{2}$

m1

$v^2 = u^2 + 2kt$

A1

[6]

Q4.

(a) $mg - 0.1m\,v = m\frac{dv}{dt}$

Newton's 2nd law

M1

$\frac{dv}{dt} = g - 0.1v$

CAO

A1

2

(b) $\int_0^v \frac{dv}{g-0.1v} = \int_0^t dt$

Attempt at integration with correct separation of variables

M1

$[-10\ln(g-0.1v)]_0^v = [t]_0^t$

–1 EE

A2,1

$\ln \frac{g-0.1v}{g} = -0.1t$

A1F

$v = 10g(1 - e^{-0.1t})$

Or equivalent

A1F

5

(c) As $t \to \infty$, $e^{-0.1t} \to 0$

M1

1

$\therefore\ v \to 10g = 98$

[8]

Q5.

(a) $1000 \frac{dv}{dt} = 2000 - 40v$

Use of Newton's 2nd law

M1

$$\frac{dv}{dt} = \frac{2000 - 40v}{1000} = \frac{50 - v}{25}$$

Correct expression from correct working

A1

2

(b) $$\int \frac{1}{50 - v}\,dv = \int \frac{1}{25}\,dt$$

$$-\ln|50 - v| = \frac{t}{25} + c$$

Integration to obtain t and 1n term

M1

Correct integration

A1

$$50 - v = Ae^{\frac{t}{25}}$$

$$v = 50 - Ae^{-\frac{t}{25}}$$

Solving for v

m1

$v = 0, t = 0 \Rightarrow A = 50$

Finding constant of integration.

m1

$$v = 50\left(1 - e^{-\frac{t}{25}}\right)$$

Correct final answer

A1

5

[7]

Q6.

(a) $m\dfrac{dv}{dt} = -mkv^3$

Forming a differential equation

M1

$$\int \frac{1}{v^3}\,dv = -\int k\,dt$$

$$-\frac{1}{2v^2} = -kt + c$$

Integrating to get a $\frac{1}{v^2}$ *term*
Correct Integral including c

m1 A1

$$v = U,\ t = 0 \Rightarrow c = -\frac{1}{2U^2}$$

Finding c
Correct c

m1 A1

$$\frac{1}{2v^2} = kt + \frac{1}{2U^2} = \frac{2ktU^2 + 1}{2U^2}$$

$$v^2 = \frac{U^2}{2ktU^2 + 1}$$

Correct final answer from correct working

A1

6

(b) v tends to zero
Allow decreases

B1

1

[7]